

DIGITAL DESIGN AND COMPUTER ARTICICTURE

4-Bit Binary Arithmetic Calculator

By Group 1

PROJECT TEAM

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ABSTRACT & OBJECTIVES

Design a 4-bit combinational arithmetic calculator

A compact logic system for arithmetic operations—built, simulated, and verified entirely in Logisim.

Core objectives



Build the arithmetic core

Process two 4-bit inputs with pure combinational logic.



Support four operations

Addition, subtraction, increment, and decrement.



Implement and verify

Construct the circuit and test behavior in Logisim.

⚡ NO CLOCK • NO MEMORY • DIRECT LOGIC

DESIGN PRINCIPLE

Every output is determined by the current inputs.

Tool Logisim

Architecture Combinational

System Overview

A compact arithmetic unit with defined inputs, outputs, and operation control.



01

Inputs

4 + 4 + 2

A, B operands plus control bits



02

Outputs

4-bit

Result bus with status flags



03

Operation Select

00–11

Add · subtract · increment · decrement

Architecture and Sub-Modules

A modular datapath turns one-bit logic into a reusable arithmetic engine with explicit status outputs.



1-Bit Full Adder

The core building block combines two operands and an incoming carry.

- +Sum and carry generated together
- +Reusable at every arithmetic bit



4-Bit Ripple-Carry

Four full adders expand the one-bit primitive into a compact arithmetic datapath.

- Carry propagates from low to high bit
- Scales arithmetic without changing the cell



Conditioning + Flags

MUX/XOR logic prepares inputs while the datapath exposes decision-ready flags.

- ◆ Carry-out and zero indicate state
- ◆ Overflow captures signed-limit errors

Truth Tables and Boolean Logic

Control inputs select the operation; the full adder turns bits into sum and carry.

01

Operation Selection

Control → output behavior

CTRL	OPERATION	RESULT
00	AND	$A \wedge B$
01	OR	$A \vee B$
10	XOR	$A \oplus B$
11	ADD	$A + B + Cin$

→ The control word is the routing decision.

02

Full Adder Behavior

Three inputs → two outputs

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

SUM

$$A \oplus B \oplus Cin$$

CARRY

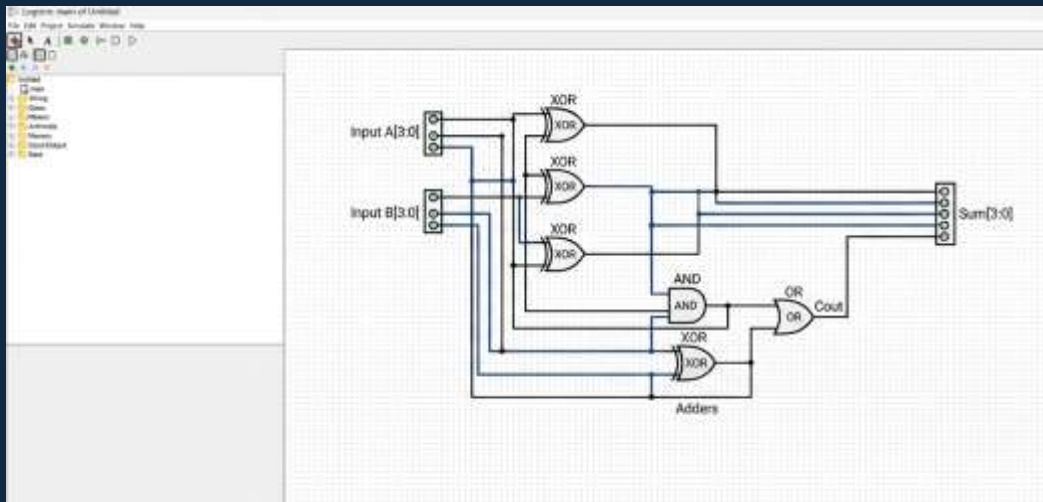
$$AB + ACin + BCin$$

Logisim Circuit Implementation

From modular logic blocks to visible signal paths, the schematic makes control flow and data flow auditable.

Logisim sample.

SCHEMATIC → SUB-CIRCUIT



● Control flow ● Active output / LED ● Inactive / blocked

Implementation notes

→ Trace the signal path

Inputs enter at the left; outputs and LEDs expose the final logic state.

📦 Reuse modular blocks

Sub-circuits isolate repeated logic and keep the top-level schematic readable.

HIERARCHY

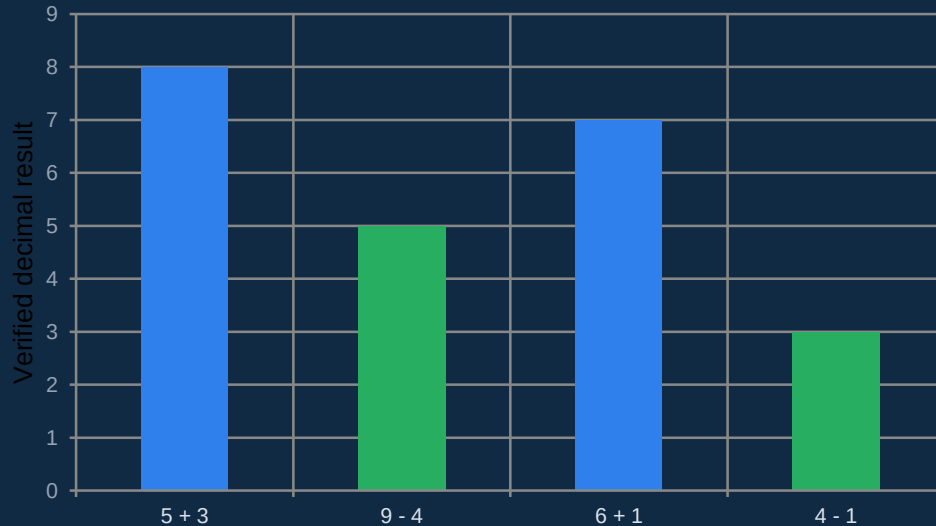
Top-level circuit → reusable sub-circuit → gates → LEDs

Simulation Results and Test Cases

Four arithmetic cases • all verified

Computed output by test case

• Result



Each bar maps directly to the stated arithmetic result.

BINARY VERIFICATION

PASSED

Outputs match expected values

Evidence in the cases

✓ Addition logic

$5 + 3 = 8$; $6 + 1 =$

7
✓ Subtraction logic

$9 - 4 = 5$; $4 - 1 = 3$

VERIFICATION INSIGHT

The simulator preserves arithmetic correctness across increment, decrement, addition, and subtraction.

Thank You...

**“A verified ALU is more than a working circuit —
it is a flexible foundation for larger arithmetic
systems.”**

Project significance



Where it applies

Core arithmetic block in processors and microcontrollers.



Why it matters

Modular combinational logic scales into larger arithmetic systems.